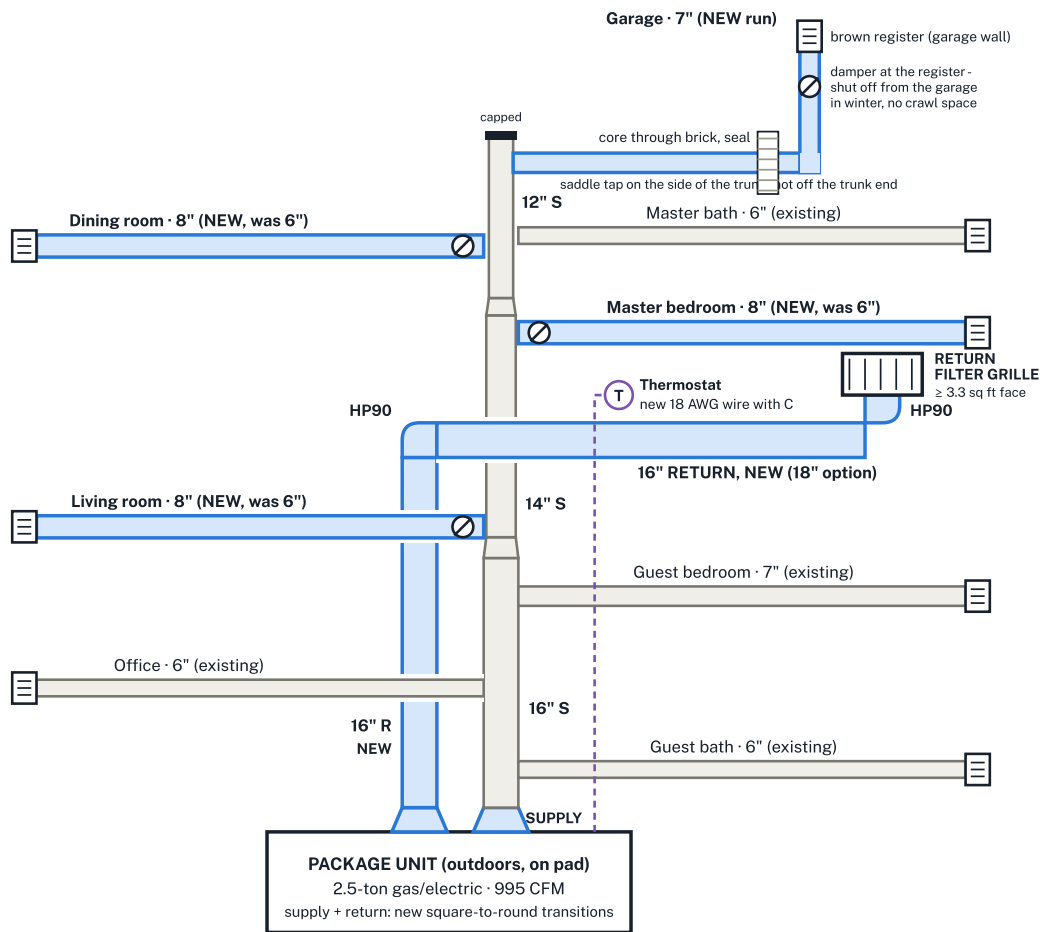




NOT TO SCALE · all sizes are round duct diameters

Duct Layout & Scope

A/C replacement · Loewen residence · design airflow **995 CFM** · not to scale

LEGEND

- New work:** replace or install
- Existing, to remain (inspect, re-seal, re-strap)
- Saddle tap with manual balancing damper
- Boot and supply register
- Hard-pipe (sheet-metal) 90° elbow
- Thermostat (owner-supplied)
- Thermostat wire (low voltage)

REQUIRED SCOPE

- Unit connections:** new square-to-round transitions on **both** supply and return at the package unit. Remove the existing twisted connections.
- Return:** new 16" insulated flex straight back from the unit, a sheet-metal 90 across the supply trunk, and a second sheet-metal 90 into the return filter grille. *16" runs about 713 ft/min at 995 CFM, the ACCA Manual D maximum for flex; quote 18" (about 563 ft/min) as an option.*
- Return filter grille:** at least 3.3 sq ft of filter face (≤ 300 ft/min), e.g. 20 × 25. Replace the existing grille if smaller.
- Upgrade 6" → 8":** Dining room, **Master bedroom**, Living room, each with a new saddle tap, balancing damper and boot.
- New 7" garage run:** saddle tap **on the side of the 12" supply trunk** (not off the trunk end); core through the brick and seal the penetration; turn up and run up the garage wall to a boot and brown register; **balancing damper at the register, inside the garage**, so it can be shut off in winter without entering the crawl space.
- Existing to remain:** supply trunk 16" → 14" → 12" with reducers; Master bath 6" (one register, adequate), Guest bedroom 7", Office 6", Guest bath 6". Inspect, re-seal and re-strap. **Replace any run with wet or damaged insulation.**
- All flex:** fully extended, no kinks, sags or compression; supported per manufacturer; every joint sealed with mastic or UL 181 tape. State the insulation R-value on the quote.
- Thermostat:** new 18 AWG thermostat wire with C-wire from unit to thermostat (no power-stealing thermostats per the Carrier 48NL manual); install owner's thermostat.
- Start-up, recorded on the invoice:** supply and return static pressure, dampers balanced, airflow set to 995 CFM, refrigerant leak-detection system tested where equipped.
- Permit pulled by the contractor; old equipment and debris removed.

PROJECT A/C replacement, duct modifications	LOCATION 301 S Aztec Dr, White House, TN 37188
EQUIPMENT 2.5-ton gas/electric package unit	DESIGN AIRFLOW 995 CFM
SOURCE Owner's field sketch, 10 Sep 2026	DRAWING Rev C · 17 Sep 2026 · not to scale